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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech (MMD) (CBCS) Degree Examinations

December 2024-January 2025
22MMD133 – MECHATRONICS SYSTEM DESIGN

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full question from Unit I and II and one full question from Unit III.

Unit – I

Marks BT* CO* PO*

1. a) Figure Q1(a) shows the reduction gear box being driven by a motor that develops a torque of $T_m(t)$. It has a gear ratio of ' n ' and the moment of inertia on the motor and output shafts are ' I_m ' and ' I_o ' and the respective damping coefficients are ' C_m ' and ' C_o '. Find the differential equation relating the motor torque ' $T_m(t)$ ' and the output angular position ' $\theta_o(t)$ '

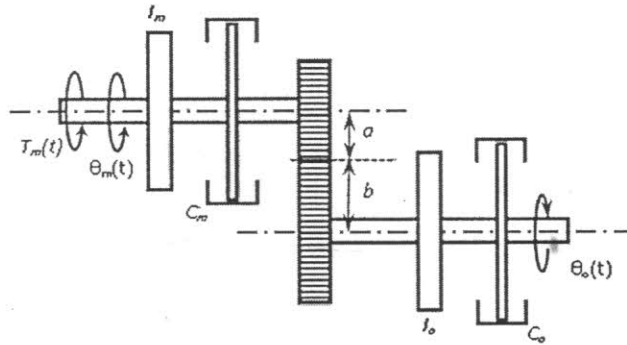


Fig. 1 (a)

- b) Explain the working of LVDT and derive the output voltage equation of LVDT.
2. a) Derive the relationship between the potential difference across the resistor capacitor and the input ' V ' for the circuit shown in the Fig. 2(a).

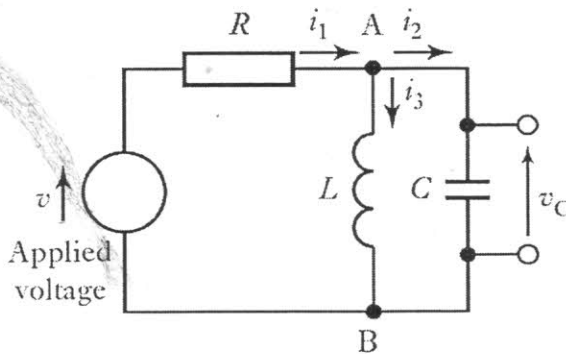


Fig. 2 (a).

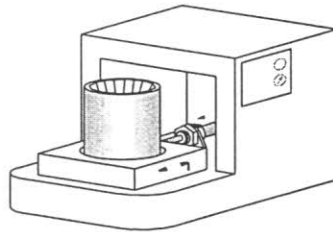
- b) Explain the working principle of absolute encoder highlighting the significance of grey code.
3. a) Obtain the equation for Hydraulic capacitance and Hydraulic inertance.
 b) Derive the equation relating the strain with the gauge factor of the strain gauge, initial resistance of the stain gauge, input voltage and output voltage of the Wheatstone bridge circuit.

10	L*3	1	1
10	L2	2	1
10	L3	1	1
10	L2	2	1

P.T.O

Unit – II

4. a) Initially the double acting cylinder is in a completely extended position. When a push button has been pressed, the extended cylinder (1A) retracts completely and executes a to-and-fro movement in the rear stroke range. The oscillating is limited to the retracted end position by a roller lever valve as well as a second roller lever valve in the central position. The frequency of oscillating is adjustable within limits by setting a pressure regulator controlling the amount of air supply. Apply an operating pressure of $p = 4 \text{ bar}$ ($= 400 \text{ kPa}$) to accomplish this to-and-fro motion. After a specified interval, the oscillator is switched off. The double-acting cylinder extends completely and actuates the third roller lever valve. Set a vibration time of $t = 5 \text{ seconds}$.



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|-------|--|----|----|---|---|
| b) | Design an active low-pass filter with a cut-off frequency of 100Hz. | 10 | L4 | 3 | 1 |
| | | 10 | L2 | 4 | 1 |
| 5. a) | Three hall-effect sensors are used to sense the position of rotor of a permanent magnet Brushless DC motor. Design a switching circuit for providing the power supply to the Coil-A, Coil-B and Coil-C of BLDC motor based on the output from the Hall-effect sensors. | 10 | L2 | 3 | 1 |
| b) | A four-bit R-2R ladder digital to analog converter has an input reference voltage of 8V. Derive the equation for the output of the R-2R ladder DAC. Obtain the analog output for all the possible combinations of 4 bit digital number. | 10 | L3 | 4 | 1 |
| 6. a) | Using a diverting device, parts are to be moved from one conveyor track onto another in a sequence. By pressing a pushbutton switch, the oscillating piston rod of a cylinder pushes the turntable via a Ratchet and Pawl mechanism in stepped sequence. The parts are diverted and transported onwards in the opposite direction. By pressing another pushbutton switch the drive unit is switched off. Design a electropneumatic Circuit for implementing this sequence. | 10 | L3 | 3 | 1 |
| b) | A four-bit weighted resistor, digital to analog converter has an input reference voltage of 1V. Assume equal resistance values for the resistor connected to LSB and the resistor connected to the feedback path of the amplifier. Obtain the analog output for all the possible combinations of 4 bit digital number. | 10 | L2 | 4 | 1 |

Unit – III

7. a) Consider a thermocouple which has a transfer function linking its voltage output V and temperature input of

$$G(s) = \frac{30 \times 10^{-6}}{10s + 1} \text{ V/}^{\circ}\text{C}$$

Determine the response of the system when subject to a step input of size 100°C and the time taken to reach 95% of the steady-state value.

- | | | | | | |
|-------|---|----|----|---|---|
| b) | Obtain the equation of natural response for a first order system. | 10 | L3 | 5 | 1 |
| | | 10 | L3 | 5 | 1 |
| 8. a) | Explain the benefits of Micro Electro Mechanical Systems. | 10 | L1 | 5 | 1 |
| b) | Explain the effect of scaling in electrostatic forces. | 10 | L1 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
